

8.1 Residential Code Quick Reference

The residential building code requirements you will reach for most often, in one place.

⚠️ VERIFY WITH YOUR LOCAL JURISDICTION

This quick reference is based on the International Residential Code (IRC). Always verify requirements with your local building department as requirements may vary and local amendments may be more restrictive than the IRC. When in doubt, consult with your building inspector before proceeding.

STAIRS, HANDRAILS & GUARDRAILS

Stairs and Handrails

Requirement	Minimum/Maximum	Notes
Maximum Riser Height	7.75 inches	Height from one tread to next
Minimum Tread Depth	10 inches	Measured horizontally
Maximum Riser Variance	3/8 inch	Between tallest/shortest risers
Maximum Tread Variance	3/8 inch	Between deepest/shallowest
Minimum Stair Width	36 inches	Clear width, not including handrails
Minimum Headroom	6 feet 8 inches	Measured vertically above nosing
Handrail Height	34-38 inches	Measured from nosing
Handrail Required	When 4+ risers	At least one side required
Handrail Graspability	1.25-2 inches	Perimeter or grip size
Handrail Clearance	1.5 inches minimum	From wall to handrail

Guardrails

Requirement	Minimum/Maximum	Notes
Required When	Drop exceeds 30 inches	Porches, decks, landings, balconies
Minimum Height	36 inches residential	Measured from deck/floor surface
Sphere Test	4 inch maximum opening	No gaps allowing 4" sphere to pass
Triangle Test	6 inch max opening	For stair guardrail triangular openings
Load Requirements	200 lbs concentrated	At top rail in any direction
Intermediate Rails	Spaced to prevent 4" sphere	Balusters, pickets, or mesh

ELECTRICAL OUTLETS AND CIRCUITS

Outlet Placement

Location	Requirement	Additional Notes
Wall Outlets - Spacing	Maximum 12 feet apart	No point more than 6 feet from outlet
Wall Outlets - Doors	Maximum 6 feet from door	Measured along wall
Kitchen Counter Outlets	Maximum 4 feet apart	Must be above counter height
Kitchen Islands	One outlet required	Island 24" x 12" or larger
Bathroom Outlets	One within 3 feet of sink	Must be GFCI protected
Garage Outlets	At least one	GFCI protected
Exterior Outlets	Front and back required	GFCI protected, weather-resistant
Basement Outlets	At least one	GFCI if unfinished
Hallways	One if 10+ feet long	Measured along centerline

GFCI PROTECTION REQUIRED

- All bathroom outlets
- Kitchen countertop outlets
- All exterior outlets
- All garage outlets
- Unfinished basement outlets
- Crawl space outlets
- Laundry room outlets (within 6 feet of sink)
- Wet bar sink outlets

SMOKE AND CO DETECTORS

Detector Placement and Power

Location	Requirement	Notes
Each Sleeping Room	One smoke detector	Inside each bedroom
Outside Sleeping Areas	One per sleeping area	In hallway near bedrooms
Each Story	At least one	Including basement, not crawl space
Interconnection	All must be interconnected	When one sounds, all sound
Power Source	Hardwired with battery backup	Or 10-year sealed battery
CO Detectors	Outside sleeping areas	Required if fuel-burning appliances
CO Detector - Each Level	One per floor minimum	Including basement

EGRESS, CEILING HEIGHTS & OTHER REQUIREMENTS

Emergency Escape and Rescue Openings (Egress Windows)

Requirement	Minimum/Maximum	Notes
Required Location	Each sleeping room	Basements with bedrooms also need egress
Minimum Net Clear Opening	5.7 square feet	Actual openable area
Minimum Opening Width	20 inches	Clear opening when open
Minimum Opening Height	24 inches	Clear opening when open
Maximum Sill Height	44 inches	From floor to bottom of opening
Window Wells (if req.)	9 sq ft minimum	If window well depth >44 inches
Window Well Width	36 inches minimum	Horizontal projection
Window Well Ladder	Required if >44" deep	Permanent ladder or steps
Ladder Requirements	Max 18" projection	Rungs 3" wide, 12" spacing max

Ceiling Heights

Room Type	Minimum Height	Notes
Habitable Rooms	7 feet	Living rooms, bedrooms, kitchens
Bathrooms	6 feet 8 inches	At fixtures and circulation paths
Hallways	6 feet 8 inches	Minimum clear height
Kitchens	6 feet 8 inches	At work areas
Basements (unfinished)	6 feet 8 inches	Beams may project to 6 feet 4 inches
Sloped Ceilings	50% of room area at min height	No portion below 5 feet
Furred Ceilings	Maintain minimum heights	Account for lowered ceiling

Additional Common Requirements

Item	Requirement
Tempered Glass	Required within 24" of door, bottom edge <60" above floor
Bathroom Ventilation	Window 3 sq ft (1.5 sq ft openable) OR mechanical fan
Mechanical Fan	Min 50 CFM intermittent or 20 CFM continuous (bathroom)
Stair Lighting	Required at top and bottom, 3-way switches if 6+ risers
Insulation	Meet energy code for climate zone (R-values vary by location)
Attic Access	Minimum 22" x 30" opening, accessible location
Crawl Space Access	Minimum 18" x 24" opening